Week 8 Report: MDP Reformulation of Marketing Campaign Budget Allocation

1. Short Reminder of the Original Model

The original project studies a marketing campaign budget allocation problem under operational constraints. The main business problem is to decide which marketing campaigns should be selected when the company has a limited marketing budget. Each campaign has an acquisition cost, expected return, audience reach, channel information, and a risk proxy.

In the current formulation, the problem is modeled as a Mixed Integer Linear Programming model. For each campaign $i \in I$, a binary decision variable and objective funcction is defined:

$$x_i = \begin{cases} 1, & \text{if campaign } i \text{ is selected} \\ 0, & \text{otherwise} \end{cases}, \qquad \max \sum_{i \in I} (RO\,I_i - \lambda Risk_i) x_i$$

where $ROI_i$is the expected return of campaign i, $Risk_i$is the risk score of campaign i, and $\lambda$is the risk penalty parameter.

The main constraints are:

$$\sum_{i \in I} Cos\,t_i x_i \leq B, \qquad \sum_{i \in I} Reac\,h_i x_i \geq R_{min}, \qquad \sum_{i \in C_k} x_i \leq L_k, \forall k \in K, \qquad x_i \in \{0,1\}, \forall i \in I$$

These constraints represent the total budget limit, minimum audience reach requirement, channel capacity limits, and binary campaign selection structure. In the static MILP model, all campaigns are selected at once. In the MDP interpretation, the same campaign selection problem is viewed as a sequential decision making process, where campaigns are evaluated one by one over a finite planning horizon.

2. Sequential Decision Interpretation

The original optimization model is static because it selects the final campaign portfolio in one step. In real marketing planning, this process can also be viewed sequentially: a manager evaluates campaigns one by one, updates the remaining budget, tracks accumulated reach, and monitors channel usage after each decision. In the MDP formulation, each campaign represents one decision stage, where the decision maker chooses whether to select or skip the current campaign. The sequence works as follows:

1. At the beginning, no campaign has been selected.
2. The available budget is equal to the initial marketing budget B.
3. The accumulated reach is zero.
4. The number of selected campaigns in each channel is zero.
5. At each stage t, one campaign is reviewed.
6. The decision maker chooses one of two actions: select the campaign or skip it.
7. If the campaign is selected, the remaining budget decreases, accumulated reach increases, and the channel count for that campaign's channel increases.

8. If the campaign is skipped, the state remains mostly unchanged except that the process moves to the next campaign.
9. The process ends after all campaigns have been reviewed.

This creates a valid sequential decision structure because each choice affects future feasibility. For example, selecting a costly campaign may limit later selections, while overusing one channel may block future campaigns from that channel. Therefore, the MDP keeps the same business logic as the MILP model but represents it as a finite horizon decision process.

3. MDP Definition

3.1 State Space S

At stage t, the state must include all information needed to make the next decision and check future feasibility, this state is sufficient because it shows which campaign is being evaluated, the remaining budget, accumulated reach, and used channel capacity :

$$s_t = (t, b_t, r_t, n_{1,t}, n_{2,t}, \ldots, n_{|K|,t})$$

Term	Meaning
t	Current campaign index / decision stage
b_t	Remaining budget before campaign t
r_t	Accumulated reach so far
$n_{k,t}$	Number of selected campaigns from channel $k$up to stage t
K	Set of marketing channels

3.2 Action Space

$$a_t \in \{0,1\}, a_t = \begin{cases} 1, & \text{select campaign } t \\ 0, & \text{skip campaign } t \end{cases}$$

Selection is feasible only if:

$$Cost_t \leq b_t, n_{k(t),t} + 1 \leq L_{k(t)}$$

Therefore:

$$A(s_t) = \begin{cases} \{0,1\}, & \text{if campaign } t \text{ is feasible} \\ \{0\}, & \text{otherwise} \end{cases}$$

Skipping is always feasible; selection requires enough budget and available channel capacity.

3.3 Transition Function

Action	Transition
Skip $a_t = 0$	$s_{t+1} = (t+1, b_t, r_t, n_{1,t}, \ldots, n_{\lvert K \rvert,t})$
Select $a_t = 1$	$b_{t+1} = b_t - Cost_t$ $r_{t+1} = r_t + Reach_t$ $n_{k(t),t+1} = n_{k(t),t} + 1$ and for all other channels $j \neq k(t)$: $n_{j,t+1} = n_{j,t}$ So the next state becomes: $s_{t+1} = (t+1, b_t - Cost_t, r_t + Reach_t, n_{1,t+1}, \ldots, n_{\lvert K \rvert,t+1})$

For other channels, $n_{j,t+1} = n_{j,t}$for $j \neq k(t)$.

In the base model, transitions are deterministic because cost, reach, ROI, and risk are known. If actual reach is uncertain, the model can be extended as:

$$Reach_t^{actual} \sim P(\cdot \mid t, a_t)$$

Then, the next accumulated reach becomes probabilistic.

3.4 Reward / Cost Function

The immediate reward follows the original risk adjusted return objective:

$$R(s_t, a_t) = (ROI_t - \lambda Risk_t)a_t$$

If the campaign is skipped, the reward is zero. To enforce the minimum reach requirement, a terminal penalty is added:

$$G(s_{T+1}) = \begin{cases} 0, & r_{T+1} \geq R_{min} \\ -M, & r_{T+1} < R_{min} \end{cases}$$

Thus, the model receives rewards from selected campaigns but is penalized if the final reach requirement is not satisfied.

3.5 Policy

$$\pi(s_t) = a_t, a_t \in A(s_t)$$

A policy decides whether to select or skip the current campaign based on budget, reach, channel usage, and risk adjusted return. A simple policy selects a campaign if it has positive risk adjusted return, fits

the remaining budget, and does not violate the channel limit. A more advanced policy can be obtained using dynamic programming or reinforcement learning.

3.6 Horizon and Terminal Condition

The problem has a finite horizon because there are Tcampaigns. The process ends at:

$$t = T + 1$$

At this point, the final portfolio is evaluated using total risk adjusted return, remaining budget, accumulated reach, channel usage, and satisfaction of R_{min}.

Property	Description
Deterministic	Campaign parameters are known in the base model
Fully observable	Budget, reach, channel usage, and campaign data are observed
Finite horizon	The number of campaigns is fixed
Undiscounted	Campaign order is only a planning sequence

$$\gamma = 1$$

4. Bellman Equation

Let $V_t(s_t)$represent the maximum total future value that can be achieved from state $s_t$at stage t. In this project, the value function means:

"The best possible remaining risk adjusted return that can be obtained from campaign $t$onward, given the current remaining budget, accumulated reach, and channel usage."

The Bellman recursion is:

$$V_t(s_t) = \max_{a_t \in A(s_t)} [(ROI_t - \lambda Risk_t)a_t + V_{t+1}(s_{t+1})] \quad for \quad t = 1,2,\ldots,T$$

The terminal value is:

$$V_{T+1}(s_{T+1}) = \begin{cases} 0, & \text{if } r_{T+1} \geq R_{min} \\ -M, & \text{if } r_{T+1} < R_{min} \end{cases}$$

The Bellman equation captures the sequential logic of the campaign allocation problem. Each campaign decision is evaluated by considering both its immediate reward and its effect on future opportunities. Each term has the following meaning:

- $V_t(s_t)$: best achievable value from the current stage to the end.
- a_t: decision to select or skip campaign t.
- $A(s_t)$: feasible actions in the current state.

- $(ROI_t - \lambda Risk_t)a_t$: immediate risk adjusted reward from selecting campaign t.
- $V_{t+1}(s_{t+1})$: best future value after the current action.
- s_{t+1}: next state after the decision.
- M: large penalty for failing to satisfy the minimum reach requirement.

5. Connection Between Original MILP and MDP Formulation

Original MILP element	MDP interpretation
Binary variable x_i	Becomes the action $a_t$at the stage where campaign $i$is evaluated. $a_t = 1$ corresponds to $x_i = 1$, and $a_t = 0$corresponds to $x_i = 0$.
Budget constraint	Embedded in the state variable $b_t$and the feasible action set $A(s_t)$. A campaign cannot be selected if its cost is larger than the remaining budget.
Minimum reach constraint	Tracked through $r_t$and enforced at the terminal condition. The final state must satisfy $r_{T+1} \geq R_{min}$, otherwise a penalty is applied.
Channel capacity constraint	Embedded through $n_{k,t}$and $A(s_t)$. A campaign cannot be selected if its channel has already reached its limit L_k.
Objective function	Converted into the sum of stage rewards. The total return of a policy equals the expected sum of $(ROI_i - \lambda Risk_i)$ over selected campaigns, adjusted by any terminal reach penalty.
Static portfolio solution	Becomes a sequential policy that decides whether to select or skip after observing the current planning state.

6. Illustrative Example

Consider a small example with two Social campaigns. Assume:

$$B = 100, R_{min} = 120, \qquad L_{Social} = 1, \lambda = 1$$

Campaign	Channel	Cost	Reach	ROI	Risk
C1	Social	60	80	20	4
C2	Social	50	70	18	2

At the first decision stage, the initial state is:

$$s_1 = (1,100,0,0)$$

This means the model is evaluating C1, with remaining budget 100, accumulated reach 0, and Social channel usage 0. Since C1 fits the budget and channel limit, the feasible actions are select or skip:

$$A(s_1) = \{0,1\}$$

If the policy selects C1, the next state becomes:

$$s_2 = (2,40,80,1)$$

and the immediate reward is:

$$R(s_1, 1) = ROI_1 - \lambda Risk_1 = 20 - 1(4) = 16$$

This action gives a positive immediate reward, but it reduces the remaining budget to 40 and uses the only Social channel slot. Therefore, at the next stage, C2 cannot be selected even though it has a positive risk adjusted reward. This example shows the main MDP logic: each decision affects not only the current reward but also future feasibility.

7. Experiment Plan for the Next Stage

The next computational study will compare the original deterministic MILP and simple sequential policies on subsets of the real marketing campaign dataset. Because the full dataset has 200,000 records, experiments will begin with sampled instances and then scale gradually.

- **What will be tested**

How campaign selection changes under different budget levels, reach targets, channel caps, and risk penalty values. A stochastic version will also test how policies perform when realized reach differs from expected reach.

- **Instance sizes**

The experiments will start with 50, 100, 250, and 500 campaign subsets. Larger instances can be added if the runtime remains reasonable.

- **Parameters and scenarios**

The model will test different values of budget B, minimum reach R_{min}, channel limits L_k, and λ, such as 0, 0.25, 0.5, 1, and 2. For stochastic tests, reach or conversion noise will be added.

- **Performance measures**

Performance will be evaluated using objective value, realized total reward, total cost, budget utilization, total reach, number of selected campaigns, channel usage, runtime, and terminal feasibility rate.

- **Baselines**

The MDP results will be compared against the original deterministic MILP, a greedy ROI per cost heuristic, a greedy risk adjusted ROI heuristic, and a random feasible selection policy.

- **Expected insight**

The MILP provides a strong static benchmark, while the MDP interpretation supports sequential policy analysis, uncertainty modeling, and future dynamic programming or reinforcement learning extensions.

8. Discussion: Benefits, Limitations, and Assumptions

The MDP formulation adds value because it makes the timing of information explicit. Instead of selecting all campaigns at once, the decision maker can update the plan as budget, reach, and channel capacity evolve. This is useful for sequential campaign planning, rolling-horizon decision making, and uncertain campaign outcomes. The Bellman recursion also creates a foundation for future algorithmic extensions such as dynamic programming, approximate dynamic programming, simulation-based policy evaluation, or reinforcement learning.

The main limitation is that a full MDP can become computationally large because the state includes budget, accumulated reach, channel counts, and stage index. To keep the formulation realistic, the report assumes a finite campaign list, observable planning variables, additive rewards, and a risk proxy derived from available campaign performance indicators. Another assumption is that campaign order is known before the planning process. Despite these simplifications, the MDP view is logically consistent with the original MILP and gives a stronger framework for future uncertainty-based experiments.